



CHADI CHAHID

PhD Student in Numerical Analysis, specializing in solving PDEs using Physics-Informed Neural Networks (PINNs)

in /chadichahid
/chahidchadi
ch.chahid@edu.umi.ac.ma
+212674583393

EDUCATION

10/2023 – present	PhD in Applied Mathematics Thesis: Parallel schemes (space/time) for solving PDEs Thesis Director/Co-Director: Prof. Lahmi Badr, Prof. Khallouq Samir Objectives: • Application of physics-informed neural networks (PINNs) as an innovative approach to solve partial differential equations. • Proof of convergence of physics-informed neural networks (PINNs) for a specific partial differential equation. • Exploitation of the domain decomposition method in conjunction with physics-informed neural networks to mitigate computational complexity.	Moulay Ismail University
09/2021 – 09/2023	Master's in Numerical Analysis and Partial Differential Equations Thesis: Schwarz-type domain decomposition methods Supervisor: Prof. Khallouq Samir	Moulay Ismail University
09/2017 – 07/2021	Bachelor's in Applied Mathematics Final Project: Solving differential equations using Laplace transform Supervisor: Prof. Mohamed Zitane	Moulay Ismail University
07/2017	Baccalaureate in Mathematical Sciences B	Moulay Ismail High School, Meknes, Morocco

EXPERIENCE

09/2021 – present	Private Professor of Mathematics High School Mathematics Professor: Teaching, Support, and Exam Preparation • Taught mathematics to 6th-year "Sciences Mathématiques" students, covering advanced topics while aligning with national curriculum standards. • Designed and delivered structured lessons, fostering logical reasoning and critical thinking skills in preparation for advanced studies. • Guided students in developing personalized study strategies, ensuring a deep understanding of the material and improved academic performance. • Regularly assessed students' progress, identified areas for improvement, and provided tailored support to address individual needs. Technologies used: PEDAGOGICAL STRATEGIESCURRICULUM ALIGNMENTADVANCED MATHEMATICSSTUDENT SUPPORTACADEMIC ASSESSMENT	Professor of Mathematics
01/2023 – present	Global Shapers Community Member • Led a project that significantly improved local community engagement and environmental sustainability as a member of the Global Shapers Community. • Organized and facilitated youth-focused workshops on entrepreneurship and leadership, reaching over 100 participants and fostering a new generation of community leaders. • Collaborated closely with local organizations to address social justice issues, leading to the implementation of several new initiatives and policies. • Implemented innovative community-driven solutions to streamline project management and enhance community impact. Technologies used: PROJECT MANAGEMENTCOMMUNITY ENGAGEMENTENVIRONMENTAL SUSTAINABILITYYOUTH EMPOWERMENT	Community Volunteer

2019 – present	Mathematics Tutor Tutor Private Mathematics Tutor: Personalized Support and Exam Preparation • Provided mathematics tutoring in specialized centers and private sessions, adapting teaching to each student's specific needs. • Prepared baccalaureate students for final exams, strengthening their understanding of key concepts and improving problem-solving techniques. • Supported students in their preparation for entrance exams to prestigious schools, developing effective study strategies and deepening advanced topics. • Assisted 4th-year middle school students in their mathematics studies, helping them consolidate basics and develop logical reasoning for upcoming years. • Developed personalized study plans, identified areas for improvement, and continuously adjusted teaching methods to optimize learning outcomes. Technologies used: DIFFERENTIATED PEDAGOGY EXAM PREPARATION ADVANCED MATHEMATICS STUDY TECHNIQUES PERFORMANCE ANALYSIS
01/2024 – 06/2024	Moulay Ismail University, Faculty of Sciences - Meknes, Morocco Teaching Assistant • Supervised tutorial sessions for first-year Analysis students focusing on Taylor Series and Limited Development. • Prepared and graded exercises to strengthen understanding of theoretical concepts. • Provided individualized support to address student challenges Technologies used: UNIVERSITY PEDAGOGY MATHEMATICAL ANALYSIS ACADEMIC SUPERVISION FORMATIVE ASSESSMENT
03/2024 – 05/2024	Internship at ALM-Solutions (Casablanca, Morocco) Intern Credit Card Fraud Detection System • Collected and prepared transaction data, ensuring it was clean and properly formatted for analysis. • Designed and trained a machine learning model adapted to credit card fraud detection. • Evaluated the model's effectiveness using metrics such as precision, recall, and F1 score. • Integrated the model into a real-time monitoring system for fraudulent transaction detection. Technologies used: MACHINE LEARNING SCIKIT-LEARN PANDAS IMBALANCED DATA TECHNIQUES COST-SENSITIVE LEARNING RANDOM FOREST MODEL
04/2022 – 2023	Part-time position at Fame-Management (Utrecht, Netherlands) Part-time Employee Website Design and Development • Prioritized user experience (UX) by designing intuitive interfaces improving usability and navigation. • Created visually appealing designs aligned with the client's brand image, incorporating modern trends and usability principles. • Ensured website responsiveness, adapting perfectly to various devices (desktops, tablets, and mobile phones) to provide a consistent user experience. • Collaborated closely with clients and stakeholders to understand requirements, integrate feedback, and provide customized solutions meeting business objectives. Technologies used: HTML5 CSS3 JAVASCRIPT BOOTSTRAP REACT.JS NODE.JS MYSQL FIGMA RESTFUL API
2018 – 03/2021	Freelance Mobile App Developer Freelance Development of an application to help immigrant children learn Moroccan Darija. • Developed engaging and culturally relevant educational materials, including vocabulary, phrases, and interactive exercises adapted for children. • Designed intuitive and child-friendly user interfaces (UI), ensuring easy navigation and a positive learning experience. • Collected user feedback and iteratively improved the application based on these insights to optimize usability and effectiveness in language learning. Technologies used: ANDROID STUDIO JAVA KOTLIN XML SQLITE FIREBASE REST API GIT GRADLE MATERIAL DESIGN JUNIT ESPRESSO

LANGUAGES

Arabic	Native language
English	Full professional proficiency
French	Full professional proficiency
Dutch	Elementary proficiency

PROUD OF

🌐 Open Source Contributions

"I am currently working on a personal project aimed at helping researchers write their articles and improve their understanding of other research work using the open-source LLaMA-3 model."

♥ Volunteer Experience

I am an active member of the Global Shapers community, which aims to contribute to the development of human capital in our cities.

COURSES & TRAINING

📄 SET Certificate (08/2021)

CERTIFICATION

Python Crash Course

CERTIFICATION

Web of Science Presence Management Certificate

CERTIFICATION

For more information, you can visit my personal website: www.chahidchadi.github.io